

Niklas Leinz

Braunschweig, Germany · niklas.leinz@tum.de
corusm.github.io · github.com/corusm · linkedin.com/in/niklasleinz

Robotics PhD researcher at TUM and Volkswagen. Works on robot foundation models for contact-rich assembly (vision-language-action policies, world models) and the teleoperation and training infrastructure to run them on real industrial robots.

EXPERIENCE

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| Doctoral Researcher (PhD)
Volkswagen AG | Aug. 2025 to Present
Wolfsburg |
| <ul style="list-style-type: none">Research robot foundation models for contact-rich assembly, including vision-language-action (VLA) policies and world models; first-author paper in preparation for ICRA 2027.Collect multimodal demonstration datasets on UR10e robots with a custom-built VR teleoperation system.Run a bimanual UR robot cell and integrate an AgiBot G2 humanoid into the same software stack.Own the pipeline end to end: data collection, policy training, validation gates, robot deployment.Evaluate AI and robotics startups as technology partners for Volkswagen's global production network. | |
| Project Engineer
Holzbau.Tech | Oct. 2024 to May 2025
Rosenheim |
| <ul style="list-style-type: none">Master's thesis: automated pick-and-place of timber elements with reinforcement learning and sim-to-real transfer.Trained DDPG+HER and PPO policies for trajectory planning with Stable-Baselines3 and Gymnasium.Set up a GPU-accelerated training pipeline with Docker and Jenkins.Supervised two working students. | |
| Platform Engineer (Dual Study)
Bosch Security and Safety Systems | Jan. 2022 to Sept. 2022
Munich |
| <ul style="list-style-type: none">Built an anomaly detection system for cloud services in PyTorch and deployed it with MLflow (Bachelor's thesis).Added server metrics in Java (Spring Boot, Micrometer) and built Grafana dashboards for monitoring. | |
| Application Engineer (Dual Study)
Bosch Energy and Building Solutions | Sept. 2019 to Dec. 2021
Düsseldorf |
| <ul style="list-style-type: none">Built a construction documentation tool used at 36 Bosch sites by over 800 employees; later ported it to Flutter.Published an open-source library of Bosch-styled Flutter components. | |

EDUCATION

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| Technical University of Munich
<i>Dr.-Ing. (PhD candidate)</i> | Munich
Aug. 2025 to exp. 2028 |
| <ul style="list-style-type: none">Thesis on foundation models for robot manipulation in automotive assembly; advisor Prof. Alois Knoll.Teaching: Simulation-Based Machine Learning in Robotics. | |
| Technical University of Munich
<i>M.Sc. Robotics, Cognition, Intelligence</i> | Munich
Oct. 2022 to May 2025 |
| <ul style="list-style-type: none">Coursework: Simulation-Based Machine Learning in Robotics, LLMs in Robotics, Machine Learning, MLOps.Exchange semester at the Technical University of Denmark, Copenhagen (autonomous systems). | |
| Baden-Württemberg Cooperative State University (DHBW)
<i>B.Sc. Information Technology</i> | Friedrichshafen
Oct. 2019 to Sept. 2022 |

SKILLS & INTERESTS

- Robot Learning:** VLA architectures, world models, imitation learning from teleoperation, reinforcement learning (PPO, DDPG+HER), impedance and torque control, sim-to-real transfer, NVIDIA Isaac Sim/Lab, ROS 2
- Machine Learning:** PyTorch (Lightning), Stable-Baselines3, MLflow, YOLO, OpenCV
- Software & Infrastructure:** Python, C#, Java, Flutter, Unity, Docker, Jenkins, Linux, Terraform, AWS, GCP
- Certifications:** KUKA Robot Programming 1 (2025), AWS Certified Cloud Practitioner (2021)
- Languages:** German (native), English (fluent), Spanish (basic)
- Interests:** TUM.ai Makeathon (2023); sailing, climbing, skiing, hiking